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WARSAW econometric **CHALLENGE**



ReSpo·Vision

Predicting Goal-Scoring Probability: An Econometric Analysis of In-Game Performance Dynamics

Group
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Abstract

This study examines whether players' in-game behavior can predict the probability of scoring later in a football match. Using data from UEFA youth competitions, we apply econometric and machine learning models, including logistic regression and XGBoost, to combine physical, technical, and contextual performance indicators.

Results show that player behavior contains meaningful predictive information, with advanced models outperforming baseline approaches. The strongest predictors are not isolated actions, but cumulative and structural features such as passing patterns, positional roles, and pressure-adjusted performance. These findings suggest that goal-scoring probability is a dynamic, context-dependent process shaped by both recent and accumulated match activity.

Keywords: goal-scoring probability, football analytics, machine learning, XGBoost, player performance, sports econometrics

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Introduction

Today's football is a game of details, where victory is increasingly determined by cool-headed analysis. Traditional coaching intuition, while still immensely valuable, is slowly becoming insufficient in a world where the pace of play increases season after season. What we once called "match feel" is now being distilled into numbers to enable faster, more accurate, and, above all, repeatable decision-making. Data has become an extra set of eyes for coaching staffs, allowing them to see what escapes the heat of battle on the pitch. [6]

However, a fundamental question arises: how can a player's impact on a match be properly assessed? Football is a particularly difficult sport for statisticians – goals are few and far between, and a single brilliant move can cover 89 minutes of an average game. Therefore, simply counting successful passes or kilometers covered is only the tip of the iceberg. The real challenge begins when we attempt to combine these two worlds: measuring how intense physical exertion affects the precision of technical decisions in the final minutes of a match.

This is precisely the focus of this work. Instead of looking at a match as a single entity, we break it down, seeking answers to what happens to a player's performance when their body is working at its peak. We are interested in the moment when fatigue begins to overwhelm technique, and how advanced mathematical models can help us predict this moment.

Understanding this relationship is not just theoretical. It provides real support for coaches, who must know when to make substitutions, and for physiotherapists, who ensure that players don't cross the fine line between peak performance and injury. By combining sprint data with modern evaluation of every action on the pitch, we strive to create a more complete, more human picture of a player, not a robot, but an athlete battling their own limitations in real time.

1 Literature Review

Micro-level analysis of football players' effectiveness constitutes an important research challenge, mainly because goals are rare, which makes it difficult to assess an individual player's impact on the match result directly. In response to this constraint, data-modeling approaches such as VAEP, proposed by Decroos et al.[1], were developed. In this framework, each on-field action - from passes and dribbles to shots - is assigned a value based on the change in the probability of scoring or conceding a goal. This approach allows a player's contribution to the game to be captured more granularly than traditional statistics do.

Another important research direction is the integration of micro- and macro-level data. Combining detailed event-level data with match aggregates enables a better understanding of the relationships between individual actions and the final match outcome. In this context, technical data such as passing characteristics and player behavior under defensive pressure are becoming increasingly important. Research on action models, particularly VAEP [1], indicates that incorporating a broad spectrum of match events and game context leads to more informative and

accurate predictions than models based solely on rare events, such as shots or goals.

A parallel area of research concerns the dynamics of physical performance and game intensity. The literature emphasizes the need to account for variation in effort over time, often by dividing the match into shorter intervals, such as 15-minute segments. Particular attention is given to high-intensity metrics, such as high-speed running (HSR) and sprints, which are considered key determinants of an athlete's physical load (Allen et al. [6]). Whiteley et al.[3] indicate that monitoring these parameters in real time can provide important information about an athlete's readiness to maintain a high level of performance.

The impact of fatigue and the distribution of effort over time on offensive performance also remains an important issue. Research examines whether cumulative effort from the start of a match, or rather short bursts of high intensity, particularly in the final stages of the match, have a greater impact on the likelihood of scoring a goal. The concept of relative intensity also emerges in this context, according to which a sudden increase in a player's activity relative to their average level in a given match may signal an increased likelihood of creating a scoring opportunity.

The importance of tactical context and external factors that moderate individual player performance cannot be overlooked and has been discussed by Memmert et al.[2]. Team formation, position on the pitch, and match status (home or away) influence the nature and number of scoring opportunities regardless of individual skill. Furthermore, lineup-management decisions, such as the timing of substitutions or a player's assigned tactical role, are crucial for interpreting player statistics.

Methodologically, the literature indicates the widespread use of advanced machine learning techniques in the analysis and prediction of sports performance [4]. Depending on the variable category, different modeling approaches are employed. Physical data, such as the number of sprints, distance, or maximum speed, are often analyzed using models such as Gradient Boosted Trees (e.g., XGBoost) or linear mixed models. Technical variables, including shot and passing quality and performance under pressure, are modeled using methods such as Random Forest or neural networks. In the case of contextual variables such as player position, formation, or match status, simpler probabilistic models, including logistic regression and Naive Bayes, are used.

In summary, building a predictive model of goal probability - for example, from data collected in UEFA youth tournaments - requires the comprehensive integration of physical, technical, and contextual information. A key challenge remains determining the extent to which observable on-field behavior can reliably predict a player's future performance. The quality of such models is most often assessed using metrics such as balanced accuracy and AUC, which capture their ability to distinguish between goal-scoring and non-goal-scoring opportunities.

2 Hypotheses

This study develops a set of hypotheses to examine empirically the relationship between players' on-field behavior and their probability of scoring a goal later in a match. These hypotheses are derived directly from the research questions and are formulated to be testable using statistical and machine-learning approaches.

H1: Players' on-field behavior during a match contains sufficient information to predict the probability of scoring a goal later in the game, with performance that significantly exceeds random chance, as measured by balanced accuracy and the area under the ROC curve (AUC).

H2: Distinct components of on-field behavior - such as offensive actions, movement intensity, and positional involvement - have a statistically significant effect on the likelihood that a player will score later in the match.

H3: Models that rely solely on sprinting activity and shot-related variables do not achieve predictive performance comparable to that of models incorporating a broader set of behavioral features.

H4: Including passing-related variables and indicators of player behavior under defensive pressure significantly improves the predictive performance of goal-scoring models relative to models based only on basic performance metrics.

H5: Short-term performance indicators (e.g., actions within the most recent 15-minute interval) exert a stronger influence on the probability of scoring a goal than cumulative performance measures aggregated over the entire match up to a given point.

H6: An increase in short-term performance intensity relative to a player's overall match-performance level is positively associated with a higher probability of scoring a goal.

H7: External factors not directly controlled by the player - such as match context, team performance, and opponent characteristics - have a statistically significant impact on the probability that a player will score, with varying levels of influence across factors.

Together, these hypotheses provide a structured framework for evaluating both the predictive value of behavioral data and the relative importance of different feature groups in determining goal-scoring outcomes.

3 Data Description

The descriptive statistics presented in the table provide a broad overview of player activity, temporal structure, and event distribution. While the dataset also includes physical performance variables such as distance and speed, these will be discussed separately due to their specific characteristics; therefore, the analysis begins with the remaining variables.

In the main dataset, the temporal design is reflected in the `checkpoint_min` variable, with a mean of 27.1 minutes and a range from 15 to 45 minutes, confirming the use of discrete 15-minute intervals. Player participation variables (`minute_in` and `minute_out`) show

substantial dispersion, indicating that the dataset includes both full-match players and those with partial playing time. This variability is expected and supports the panel structure of the data.

Short-term activity variables (`last15_`) provide insight into player behavior within recent time windows. In terms of counts, these variables appear plausible and informative. For example, players perform on average 1.49 sprints and 6.45 high-speed runs in a 15-minute interval. Shooting-related variables within this group are highly sparse: the mean number of shots is 0.168, while shots on target average only 0.061, with medians equal to zero across all shooting metrics. This indicates that most time intervals do not contain any shooting activity, which is consistent with the nature of football matches.

Cumulative variables (`cumul_`) capture aggregated player activity up to a given checkpoint. On average, players accumulate 4.17 sprints and 18.14 high-speed runs, reflecting differences in overall involvement and workload throughout the match. Similarly to the short-term variables, cumulative shooting statistics remain low, reinforcing the rarity of offensive events at the individual level.

The dependent variable, `scored_after`, further reflects this sparsity, with a mean of 0.058. Formally, `scored_after=1` if the same player scores later in the remaining match time after a given checkpoint, and 0 otherwise. This means that only about 5.8% of observations correspond to a later goal by that player, resulting in a highly imbalanced classification problem. Such imbalance has important implications for modeling, particularly in terms of evaluation metrics and the need for techniques that account for rare events.

The supplementary datasets provide additional context for player actions. Passing events are distributed relatively evenly over time, with a mean occurrence around the 23rd minute, suggesting consistent involvement in ball circulation throughout the match. Events recorded under pressure show a wide range of pass angles, indicating diverse directional decisions when players are constrained by opponents. Running-related events also follow a similar temporal pattern, with observations concentrated around the midpoints of match halves. Finally, shot events occur on average around the 25th minute, aligning with the general distribution of other event types.

The primary limitation of the dataset is the improper estimation of player distances derived from computer vision methods. The recorded values include implausibly large distances for a substantial subset of players, which introduces significant bias into this variable and all features derived from it.

For context, the current 10 km world record, set by Joshua Cheptegei, is 26:11 (1571 seconds), corresponding to an average speed of approximately 6.36 m/s. Even under the unrealistic assumption that a player could sustain this world-record pace continuously for 15 minutes (900 seconds), the maximum achievable distance would be:

$$\text{Distance} = 6.36 \times 900 \approx 5,724 \text{ meters.}$$

However, the dataset contains values exceeding 8 km within the same time interval, which

is physiologically implausible. Moreover, approximately 30 players in the dataset are reported to have covered distances greater than 5.5 km in 15 minutes, further highlighting the scale of the distortion.

These inconsistencies strongly suggest that the distance variable is severely biased. Consequently, all features derived from it - such as maximum speed (with recorded values exceeding 50 km/h), peak speed, and their cumulative counterparts - are also unreliable and should not be considered valid for modeling purposes.

Two key conclusions follow from this observation. First, distance-based variables must be excluded from the analysis, as their inclusion would likely lead to misleading inferences regarding feature importance and model behavior. Second, the issue points to potential technical deficiencies in the configuration or application of the computer vision system used for data acquisition (e.g., tracking inaccuracies, calibration errors, or identity switching), which warrants further investigation.

To complement the player-quarter level description, Tables 1–4 summarize the main player-level and checkpoint-level patterns, while Tables 5–10 describe the structure of the run-level dataset, including distance and speed measures, the distribution of run types, variation across match periods and pitch zones, and the possession sequences with the highest run intensity.

Table 1. Descriptive statistics

	mean	50%	std	min	max
last15_sprints	1.490820	1.000	1.421015	0.0	8.00
last15_hsr	6.447504	6.000	3.836168	0.0	20.00
last15_distance	367.125017	107.195	978.769500	0.0	8028.31
last15_mean_max_speed	6.217783	6.638	1.760838	0.0	14.14
last15_peak_speed	7.278741	7.720	2.184753	0.0	14.14
last15_shots	0.167814	0.000	0.428815	0.0	4.00
last15_shots_on_target	0.060815	0.000	0.258629	0.0	3.00
last15_shots_under_press	0.131383	0.000	0.375668	0.0	3.00
last15_shots_top_third	0.162077	0.000	0.421593	0.0	4.00
cumul_sprints	4.171543	3.000	3.681721	0.0	22.00
cumul_hsr	18.144291	16.000	12.880450	0.0	80.00
cumul_distance	969.208127	277.835	2701.188546	0.0	25667.77
cumul_mean_max_speed	6.471886	6.677	1.298648	0.0	12.02
cumul_peak_speed	8.045838	8.310	1.814146	0.0	14.14
cumul_shots	0.428285	0.000	0.799493	0.0	6.00
cumul_shots_on_target	0.155766	0.000	0.435978	0.0	4.00
cumul_shots_under_press	0.335628	0.000	0.678278	0.0	4.00
cumul_shots_top_third	0.418531	0.000	0.784418	0.0	5.00

Table 2. Percentage distribution of target by checkpoint

checkpoint	0	1
ET1_15	97.619048	2.380952
H1_15	91.042584	8.957416
H1_30	92.941176	7.058824
H1_45	94.264706	5.735294
H2_15	96.029412	3.970588
H2_30	96.323529	3.676471
H2_45	95.348837	4.651163

Table 3. Chance of scoring by position

position	scored_after (%)
A	12.520325
M	6.732976
D	3.049759
G	0.000000

Table 4. Average physical activity by position

position	last15_sprints	last15_distance
A	2.071545	476.234065
D	1.430177	390.076437
G	0.069182	12.426195
M	1.621270	380.204353

Table 5. Descriptive statistics of runs

	mean	50%	std	min	max
distance	47.927390	12.074841	121.727855	2.892288	1694.008
min_speed	5.351574	5.510000	0.895170	0.290000	7.280
max_speed	6.717121	6.720000	0.798054	5.440000	14.140

Table 6. Percentage distribution of run types

run_type	proportion (%)
hsr	80.884069
sprint	19.115931

Table 7. Average speed and distance by run type

run_type	distance	min_speed	max_speed
hsr	43.165991	5.073733	6.438401
sprint	68.074003	6.527184	7.896452

Table 8. Number of runs by match period and run type

period	hsr	sprint
extra_time_1	252	43
extra_time_2	221	53
half_1	13879	3166
half_2	14065	3454

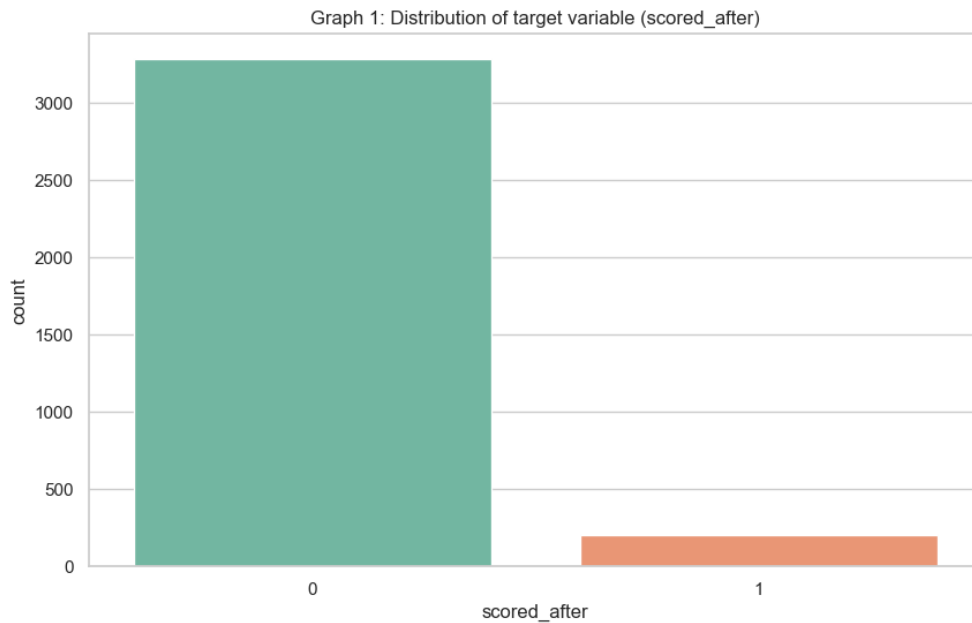
Table 9. Average run distance and max speed by pitch zone

stage	distance	max_speed
bottom	47.874102	6.698353
middle	49.086553	6.711082
top	46.359545	6.742402

Table 10. Top 5 possession sequences with the highest volume of runs

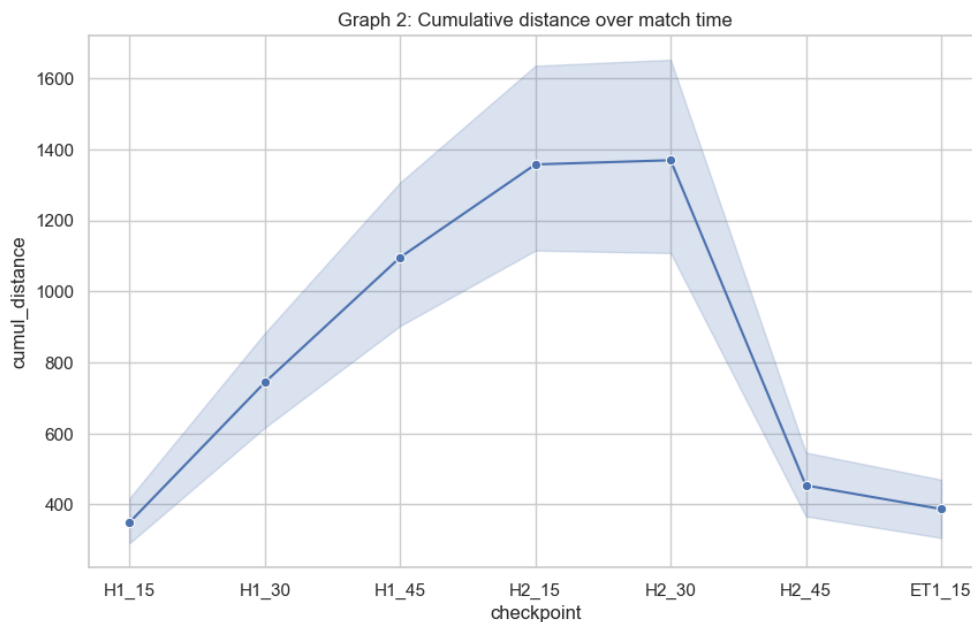
possession	runs
2377	801
2299	778
2348	748
2434	733
3753	714

The following figures complement the descriptive tables and provide a visual summary of the uploaded graph outputs.



Graph 1. Distribution of target variable.

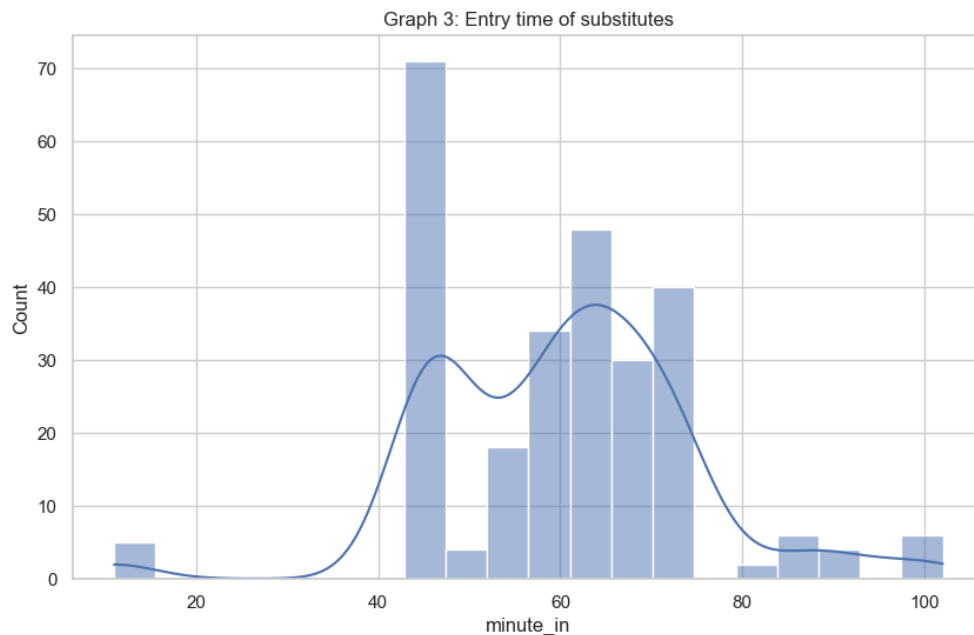
The severe imbalance of the classification problem is immediately evident in Graph 1, where the mean of `scored_after` equal to 0.058 manifests as a dominant "0" class. This visual confirmation aligns with the established finding that only 5.8% of observations result in a subsequent goal.



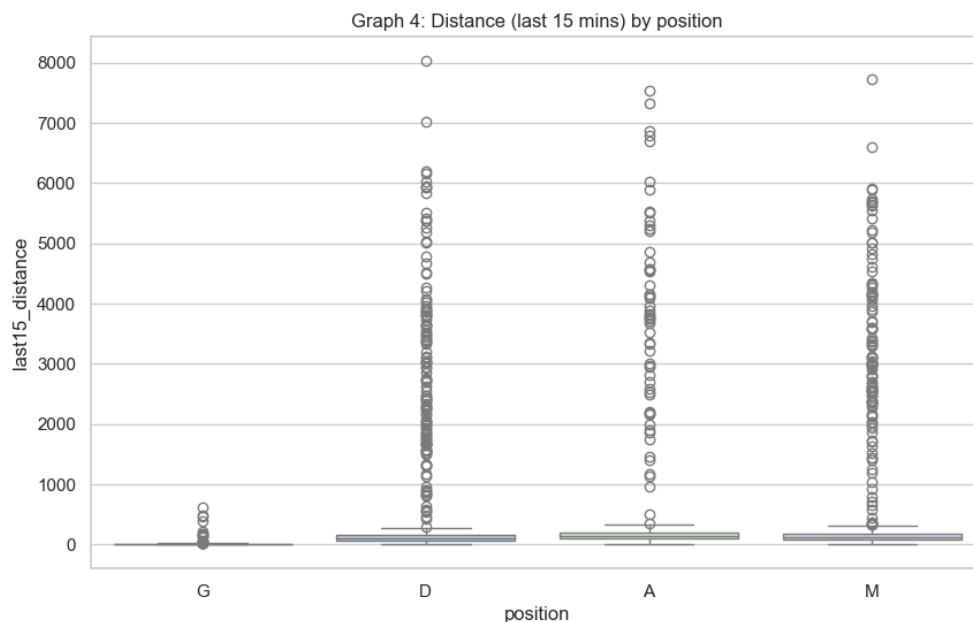
Graph 2. Cumulative distance over match time.

The "substantial dispersion" noted in player participation variables (`minute_in`, `minute_out`) is visually substantiated by Graph 3. The peaks at minute 45 and the 60–75 minute window confirm the entry of substitutes. This temporal fluctuation directly explains the

"dip" in Graph 2 for late-match checkpoints; the mean cumulative distance drops because the population at H2_45 and ET1_15 is heavily weighted by fresh substitutes who entered with zero prior activity, dragging down the group average



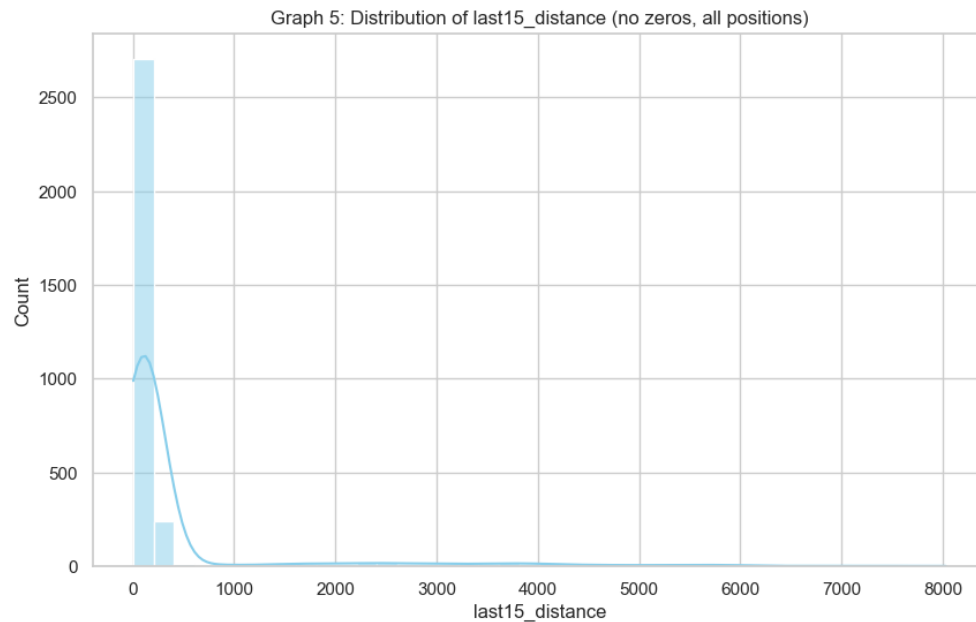
Graph 3. Entry time of substitutes.



Graph 4. Distance (last 15 mins) by position.

While Graph 4 confirms expected positional profiles - showing that midfielders and attackers typically record higher exertion than defenders and goalkeepers - it also highlights the "improper estimation" of player distances. The graph displays extreme outliers exceeding 8,000 meters in a single 15-minute interval. When measured against the physiological benchmark

of the 10 km world record pace, which would allow a maximum of roughly 5,724 meters in the same timeframe, these values are clearly impossible. This visual evidence confirms that distance-based variables and their derivatives, such as peak and maximum speed, are severely biased by tracking inaccuracies or identity switching and should be excluded to prevent misleading model inferences.



Graph 5. Concentration of runs across possession sequences.

4 Data Preprocessing

The data preprocessing pipeline integrated five contest datasets through a unified intake and normalization framework to ensure consistency across all sources prior to feature engineering. Each dataset was loaded in two parallel forms: a raw version, in which missing-value tokens were preserved for diagnostic inspection, and a normalized version, in which standardized missing-value handling was applied using a predefined set of tokens (e.g., “NA”, “N/A”, “null”, “nan”). This dual representation allowed both auditing of original data quality issues and construction of a clean analytical dataset for downstream processing. All transformations were aligned with a centralized intake module to ensure reproducibility and consistency across datasets.

Initial exploratory analysis involved the construction of dataset-level summaries capturing row counts, column counts, proportions of missing values, and duplicate observations. In addition, a global schema mapping was generated to assess column overlap across datasets, supporting future decisions regarding feature alignment and potential joins. Missing data were further examined at two levels: first, through a raw token scan identifying non-standard encodings of missingness in the original data, and second, through a normalized column-wise

summary quantifying missing proportions, non-null counts, and feature cardinality. These steps provided a comprehensive understanding of data completeness and structure prior to feature construction.

A major transformation focused on temporal representation within the player-quarter dataset. Match time variables (`minute_in` and `minute_out`) were discretized into interpretable checkpoint windows corresponding to meaningful phases of a football match, including 15-minute intervals across regulation time and extended intervals for extra time. Initially, binary indicator variables were created for each checkpoint window to represent the presence of events within specific temporal segments. To reduce dimensionality and improve interpretability, these binary indicators were subsequently compressed into categorical variables representing the active checkpoint window for each observation. During this process, redundant raw time variables and intermediate checkpoint features were removed. In addition, a derived feature representing playing duration within an appearance was constructed as the difference between entry and exit minutes plus one, providing a compact measure of participation time.

Event-level pass data were then integrated into this temporal framework. Match event timestamps were converted into absolute match minutes using period-based offsets for first half, second half, and extra time. These absolute times were mapped onto the previously defined checkpoint windows, enabling alignment between event data and player-quarter structure. Within each checkpoint, pass-related statistics were aggregated at the player-appearance level, including counts of passes attempted, successful passes, passes received, and successful received passes. These features were computed separately for passing and receiving roles and merged into the main player-quarter dataset. Missing values resulting from the join operation were imputed as zeros, and all aggregated features were cast to integer format for consistency.

To capture temporal dynamics across the progression of a match, cumulative features were then derived by ordering checkpoint observations for each player and computing running sums of pass-related metrics. This resulted in cumulative versions of passes attempted, successful passes, passes received, and successful received passes, which reflect the evolution of player involvement over time rather than isolated interval activity.

The final output of this pipeline is a structured, model-ready dataset that integrates normalized temporal representations, engineered duration features, checkpoint-aligned event statistics, and cumulative interaction measures. All intermediate temporal indicators and redundant raw variables were removed to ensure compactness and reduce multicollinearity. The resulting dataset, exported as `players_quarters_final_step1.csv`, constitutes the foundational feature set for subsequent feature selection and predictive modeling stages.

5 Theoretical Framework

5.1 Pooled Panel Logistic Regression

In this study, we employ a Pooled Panel Logistic Regression model. Unlike fixed or random effects models, the pooled approach treats each player-checkpoint observation as a distinct data point while assuming the relationship between predictors and the outcome is constant across the sample.

The probability of player i scoring after interval t in match m is given by:

$$P(y_{itm} = 1 | \mathbf{x}_{itm}) = \Lambda(\beta^\top \mathbf{x}_{itm}) = \frac{1}{1 + \exp(-\beta^\top \mathbf{x}_{itm})} \quad (1)$$

To account for within-match dependence in the data, we employ **Clustered Standard Errors** at the match level, with clusters defined by `fixture_id`. This allows observations from the same fixture to share common shocks related to tempo, tactics, and game state.

Components and Mechanics:

- $\mathbf{x}_{itm}$: The vector of inputs, including short-term (`last15_`) and cumulative (`cumul_`) metrics for player i during match m .
- **Independence Assumption:** We assume $E[\epsilon_{itm} | \mathbf{x}_{itm}] = 0$. While observations are assumed to be independent across different fixture clusters, the error terms may be correlated within the same fixture because multiple player-checkpoint observations share the same match environment.
- **Variance-Covariance Matrix:** The clustered sandwich estimator can be written as:

$$\hat{V}_{clustered} = H^{-1} \left(\sum_{g=1}^G s_g s_g^\top \right) H^{-1} \quad (2)$$

where g indexes a fixture cluster, s_g is the cluster-level score contribution, and H is the observed Hessian (equivalently, the bread matrix for the fitted model).

Application to Football Data:

- This approach is particularly relevant for the UEFA Under-21 dataset, as it ensures that the statistical significance of performance metrics (like sprints or passes) is not inflated by the repeated nature of the 15-minute checkpoints.
- It allows for a direct comparison of short-term performance intensity relative to overall performance level by treating the coefficients as global effects across all players.

5.2 eXtreme Gradient Boosting (XGBoost)

To address Research Questions regarding non-linear interactions between physical output and technical actions, we utilize XGBoost. It builds an ensemble of trees to minimize a regularized objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (3)$$

Where the complexity term Ω prevents overfitting:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda ||w||^2 \quad (4)$$

Application to Football Data:

- This model handles the high-dimensional space created by merging event-level data (e.g., behavior under pressure) with aggregated checkpoint data.
- It captures "micro-level actions" that lead to "macro-level outcomes" without assuming linear relationships [1].

5.3 SHAP (SHapley Additive exPlanations)

SHAP is used to explain the output of the XGBoost model by attributing the difference between the actual prediction and the average prediction to each feature. The explanation model is an additive feature attribution method:

$$f(x) = g(z') = \phi_0 + \sum_{j=1}^M \phi_j z'_j \quad (5)$$

Application and Components:

- ϕ_0 : The base value, representing the mean goal-scoring probability across the training set.
- ϕ_j : The SHAP value for feature j , quantifying its contribution.
- z'_j : A binary variable indicating the presence of a feature.

Global Interpretability:

Aggregated SHAP values help determine if short-term intensity or external contextual factors matter most for scoring.

5.4 k-Nearest Neighbors (k-NN)

The k-Nearest Neighbors algorithm is a non-parametric supervised learning method used for classification. Unlike the previously discussed models, k-NN does not explicitly learn a fixed set of parameters but instead makes predictions based on the local similarity of data points in the feature space.

For a given query point x_0 (a specific player-checkpoint), the model identifies the k closest observations in the training set $\mathcal{D}$ based on a distance metric $d(x_0, x_i)$. The predicted probability of scoring is calculated as the proportion of these neighbors that resulted in a goal:

$$\hat{P}(y = 1|x_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_0} I(y_i = 1) \quad (6)$$

Where:

- $\mathcal{N}_0$: The set of k points in the training data closest to x_0 .
- $I(\cdot)$: An indicator function that equals 1 if the neighbor i scored after that interval and 0 otherwise.

The distance is typically measured using the **Euclidean distance**:

$$d(x, y) = \sqrt{\sum_{j=1}^M (x_j - y_j)^2} \quad (7)$$

Application to Football Data:

- **Feature Scaling:** Since football metrics vary significantly in units (e.g., `distance_covered` in meters vs. `shots` as a count), features must be normalized (standardized) to ensure that high-magnitude variables do not dominate the distance calculation.
- **Similarity Mapping:** This model evaluates whether similar on-field profiles—such as players with high `last15_sprints` and many `last15_passes`—tend to result in goals regardless of the specific player's identity.
- **Computational Note:** While k-NN is intuitive, its performance may be limited by the high-dimensional nature of the event-level data (the "curse of dimensionality") compared to the regularized approach of XGBoost.

6 Model Validation Framework

The data splitting and validation strategy was designed to strictly prevent information leakage while preserving representative structure across matches, teams, and players. First, a

final holdout set was created using a grouped split based on `fixture_id`, ensuring that no single fixture appears in both the development and holdout datasets. The assignment to the holdout set was carefully balanced with respect to target prevalence, as well as team and player participation. In cases where explicit team identifiers were not available within the modeling table, a proxy definition of team identity was employed (e.g., fixture side constructed from `fixture_id` combined with a home/away indicator), and balance was verified using diagnostics based on player composition. All model development, including training and validation procedures, was conducted exclusively on the development split.

Within the development dataset, model evaluation relied on grouped cross-validation, again using `fixture_id` as the grouping variable to eliminate cross-match leakage. A custom fold assignment procedure was implemented to jointly balance target prevalence and ensure adequate representation of teams and players across folds. Stratification by the target variable distribution was enforced within this grouped setup to maintain consistent class proportions. Additionally, explicit balance tolerances were defined and monitored for each fold, covering both target prevalence and the distribution of team and player participation. To further safeguard against leakage, all preprocessing steps and feature engineering transformations were performed strictly within each training fold and then applied to the corresponding validation fold. This fold-specific preprocessing pipeline prevents contamination between training and validation data and was complemented by separate controls ensuring temporal causality within matches.

7 Results

7.1 Baseline Model: Pooled Logistic Regression

As a benchmark for more advanced machine learning algorithms, a baseline model based on classical econometrics was constructed. Given that the dependent variable is binary, the data has a panel structure, and there is a high concentration of zeros (rare events), a Pooled Logistic Regression is a natural choice. It estimates the probability of a player scoring in the remaining match time by treating observations as cross-sectional, while properly correcting standard errors.

Specification and General-to-Specific Approach To select the most relevant explanatory variables, the classical General-to-Specific methodology was applied. The modeling process began with a highly parameterized general model containing a broad set of variables. Dimensionality was then reduced using backward stepwise selection based on the Akaike Information Criterion (AIC). This process yielded a parsimonious model that retains high informational value with a minimal number of parameters. To account for the unobserved panel structure and the interdependence among players participating in the same match, Clustered Robust Standard

Errors at the match-player level (`player_appearance_id`) were applied. This adjustment makes the estimates robust to intra-cluster error autocorrelation.

Table 11. Goal-Scoring Probability: Pooled Logit

	Dependent variable: Scored After (1=Yes)
<code>positionD</code>	-1.197** (0.453)
<code>positionM</code>	-0.573 (0.316)
<code>is_home</code>	-0.217***
<code>last15_shots</code>	0.726* (0.361)
<code>last15_hsr</code>	0.004 (0.027)
<code>last15_pass_received</code>	-0.150 (0.077)
<code>last15_pass_received_accurate</code>	0.169 (0.093)
<code>history_shots</code>	0.411*** (0.458)
<code>history_shots_under_press</code>	-0.457 (0.457)
<code>history_hsr</code>	-0.039 (0.017)
<code>history_pass_passed_accurate</code>	-0.085***
<code>history_pass_received</code>	-0.100
<code>history_pass_received_accurate</code>	0.191** (0.069)
<code>last15_shots:last15_hsr</code>	-0.082 (0.044)
Constant	-1.566** (0.546)
Observations	3,168
Log Likelihood	-695.456
Akaike Inf. Crit.	1,420.911

Note:

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

`history_` variables represent cumulative match statistics prior to the last 15 minutes.

Estimation Results and Interpretation An analysis of the odds ratios for selected variables provides interesting analytical insights into the profile of a potential goalscorer: Pitch role: As expected, being a defender or a midfielder drastically reduces the chance of scoring compared to forwards (the baseline). The OR for `positionD` is approximately 0.302 (a nearly 70% decrease in odds). Short-term offensive activity: The strongest positive predictor in the short time window is taking shots (`last15_shots`). An increase in this metric significantly raises the odds of scoring, confirming that a player who is offensively active in recent minutes maintains a high probability of scoring. Receiving vs. Passing: The `history_pass_received_accurate` variable exhibits an OR of 1.210, meaning each accurately received pass in the match history increases a player's odds of scoring by 21%. Conversely, `history_pass_passed_accurate` shows an inverse relationship (OR = 0.919). This implies that players who frequently and accurately distribute the ball are less likely to find themselves in scoring positions.

Predictive Power and Balanced Accuracy Due to the severe class imbalance in the dataset (scoring a goal after a specific checkpoint is a rare event), evaluating the model's predictive capability required the use of Balanced Accuracy. Unlike standard Accuracy, which inherently

favors the majority class, Balanced Accuracy is the arithmetic mean of Sensitivity (true positive rate) and Specificity (true negative rate). For the cutoff of 0.058, the baseline econometric model achieved a Balanced Accuracy of 60.02% on the test set. This result confirms that a classical logistic model can successfully identify non-random patterns in the data, establishing a solid benchmark for more complex, tree-based models.

7.2 V6 XGBoost Poisson Modeling Results

The V6 XGBoost Poisson line represents the final tree-based modeling phase built on the most advanced version of the engineered feature space. Its role in the paper is broader than that of a single high-performing benchmark. First, it provides a final structured test of whether a Poisson-style intensity formulation is well suited to the rare-event problem of later goal scoring. Second, it serves as a feature-space audit: by comparing the full V6 representation with SHAP-selected reduced subsets, it makes it possible to assess whether the final engineering pipeline had reached a point of diminishing returns or even feature-induced noise. Third, because the model was optimized for balanced accuracy and then carefully calibrated, the V6 phase offers one of the clearest bridges between ranking performance, thresholded discrimination, and probability quality.

7.2.1 Dataset and Feature Composition

The V6 dataset is the result of a multi-stage feature-engineering pipeline built around match-state checkpoints from the U21 EURO 2025 analytical setup. Conceptually, each observation represents a snapshot of the match at a specific checkpoint, with the target variable `scored_after` indicating whether that player scored later in the remaining match time after the checkpoint. The full V6 representation contains 128 active model features and is explicitly designed to combine count-based physical-load indicators, event structure, tactical context, and intensity normalization into a single high-dimensional prediction space.

The data were partitioned into a development set of 2,830 observations and a strictly independent holdout set of 656 observations. Within the development data, model tuning relied on grouped and stratified five-fold cross-validation. Grouping was defined at the fixture level, so all observations from the same match were assigned to the same fold. This is methodologically important because it prevents temporal and contextual leakage between snapshots of the same game. At the same time, stratification on the `scored_after` target preserved a stable rare-event distribution across folds, while the grouping logic naturally limited overfitting to repeated player or team signatures appearing within the same fixture.

The feature space itself was layered. The earliest layer consisted of count-based physical activity summaries, especially sprinting and high-speed-running frequency together with exposure-normalized workload measures. Distance-derived and speed-derived variables—including distance, peak speed, and pace-delta features—were excluded from the final V6

representation because the tracking distortion documented earlier made them unreliable. A second layer added cumulative and last-15-minute event aggregations for passes, pressures, and shots. A third layer introduced tactical and contextual information, such as parsed formation structure, substitute or fresh-legs indicators, time-on-pitch normalization, and team-level momentum summaries. A fourth layer contributed more advanced proxies, most notably a simplified threat measure and pass-centrality style ratios. Finally, the V5/V6 stage added exposure normalization, encoding the time remaining on the pitch and allowing the model to treat later scoring as an intensity process rather than as a flat binary event detached from exposure length.

7.2.2 Modeling Strategy

The V6 modeling strategy was designed specifically for a rare-event football prediction problem. The core estimator was an `xgboost.XGBRegressor` with the `count:poisson` objective. Instead of using a standard binary objective, the model treated goal occurrence as the realization of an underlying Poisson intensity process. This is a substantively appealing choice because later scoring is infrequent, and the underlying question is not only whether a goal occurs, but how intense the scoring risk is at a given checkpoint.

Exposure was incorporated directly into the base margin through the log of the exposure variable. In practice, this means the model learned a scoring intensity per unit of remaining opportunity rather than a probability detached from playing time. After fitting, predicted intensities were transformed into event probabilities through the standard Poisson mapping $P(Y \geq 1) = 1 - e^{-\lambda}$. This allowed the model to remain faithful to the rare-event intensity logic while still being evaluated with conventional classification metrics.

Hyperparameter optimization was carried out with Optuna using the Tree-structured Parzen Estimator sampler with multivariate search enabled. The search space covered estimator count, depth, learning rate, child-weight control, subsampling, column subsampling, and regularization strengths. The primary objective for optimization was balanced accuracy. This choice was deliberate: because the target is strongly imbalanced, naive optimization of overall accuracy would almost certainly favor the majority non-scoring class. Balanced accuracy instead forces the model to maintain meaningful sensitivity to the rare positive class while preserving specificity on the negatives.

7.2.3 Calibration and Evaluation Framework

Because the raw output of the V6 model is an intensity score rather than a directly calibrated class probability, probability calibration was an essential part of the evaluation framework. The study therefore tested several post-hoc calibration methods, including isotonic regression, sigmoid scaling, beta calibration, and spline calibration. Calibration quality was assessed using the Brier score, which is especially appropriate here because it measures the

squared discrepancy between predicted probabilities and observed outcomes and therefore directly captures probability reliability.

The main evaluation metrics for the V6 line combined thresholded and ranking perspectives. Balanced accuracy served as the primary optimization objective and the main classification-oriented comparison metric. PR AUC was used to assess ranking quality in the rare-event regime, where early retrieval of positives matters more than overall ROC geometry alone. ROC AUC remained useful as a general separability measure, while the Brier score captured calibration quality before and after adjustment. This multi-metric perspective is important for the paper because the V6 line is not only about finding a strong classifier; it is also about understanding which specification best balances discrimination, ranking, and usable probability output.

7.2.4 Feature Importance and SHAP-Guided Reduction

The full V6 model was interpreted with TreeExplainer SHAP values, and this analysis revealed that predictive importance was concentrated in a relatively small subset of variables. The top twenty features included cumulative passing shares, cumulative threat, cumulative in-out passing ratio, pressure-adjusted passing accuracy, positional information, cumulative pressure structure, team shooting context, selected count-based physical-activity summaries, and positional dummy variables. The ranking was led by variables such as `cumul_pass_top_share`, `cumul_pass_middle_share`, `cumul_threat`, `cumul_in_out_ratio`, and `cumul_pressure_accurate_rate`, alongside role indicators such as `position`, `is_pos_D`, and `is_pos_A`.

This ranking is substantively revealing. It suggests that the strongest signal in the V6 space is not dominated by isolated shot counts or by the retained count-based physical indicators alone. Instead, the most influential predictors describe territorial passing structure, accumulated threat, pressure-sensitive passing quality, and role-dependent context. In football terms, later scoring risk appears to depend heavily on how possession is distributed, how threat accumulates over time, and how a player functions within the broader tactical structure of the match.

The SHAP ranking was then used to define reduced feature subsets containing the top 10, top 20, and top 30 predictors. The purpose of this reduction was to test whether the full 128-feature space was already too broad and whether the model would become stronger when restricted to the most credible signals. In other words, the ablation study directly examined whether the mature V6 feature space still contained useful incremental information, or whether it had begun to accumulate redundant and noisy variables that weakened the model.

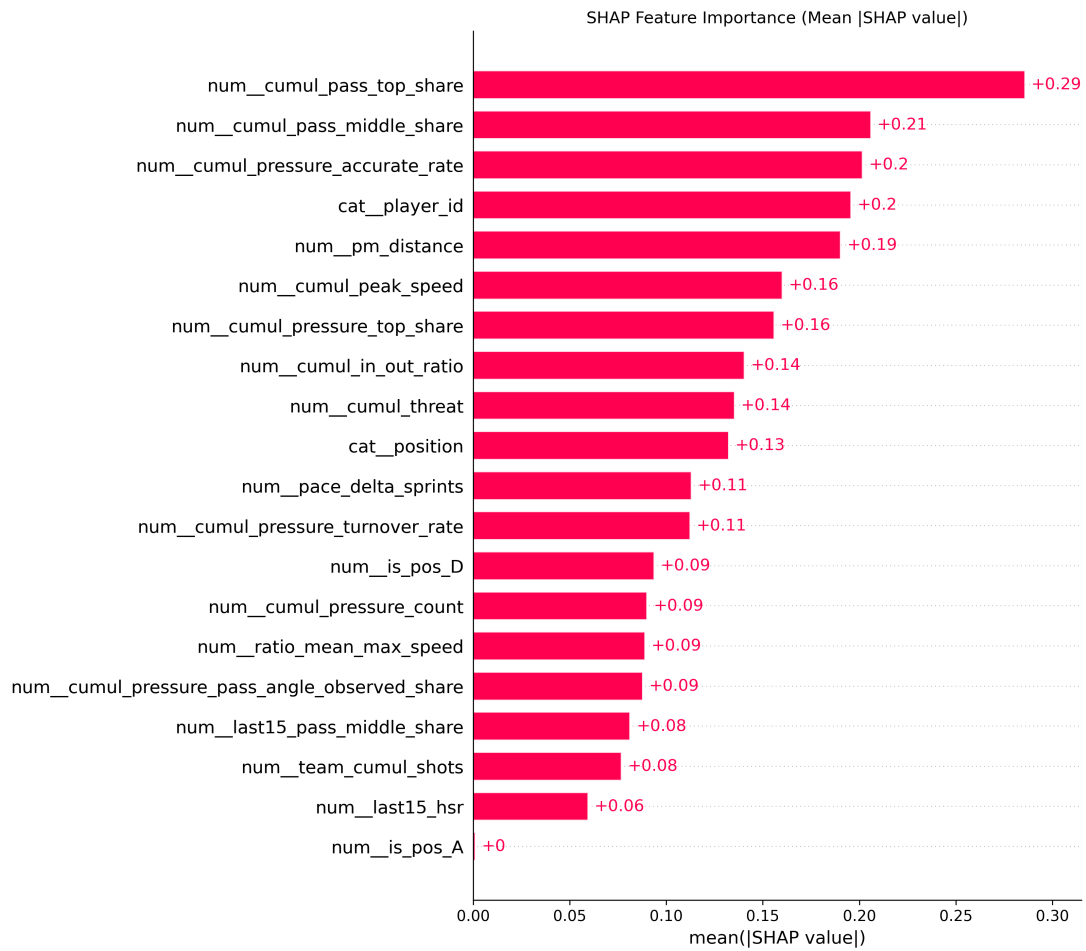


Figure 1. Top-20 SHAP feature importance for the final V6 XGBoost Poisson model.

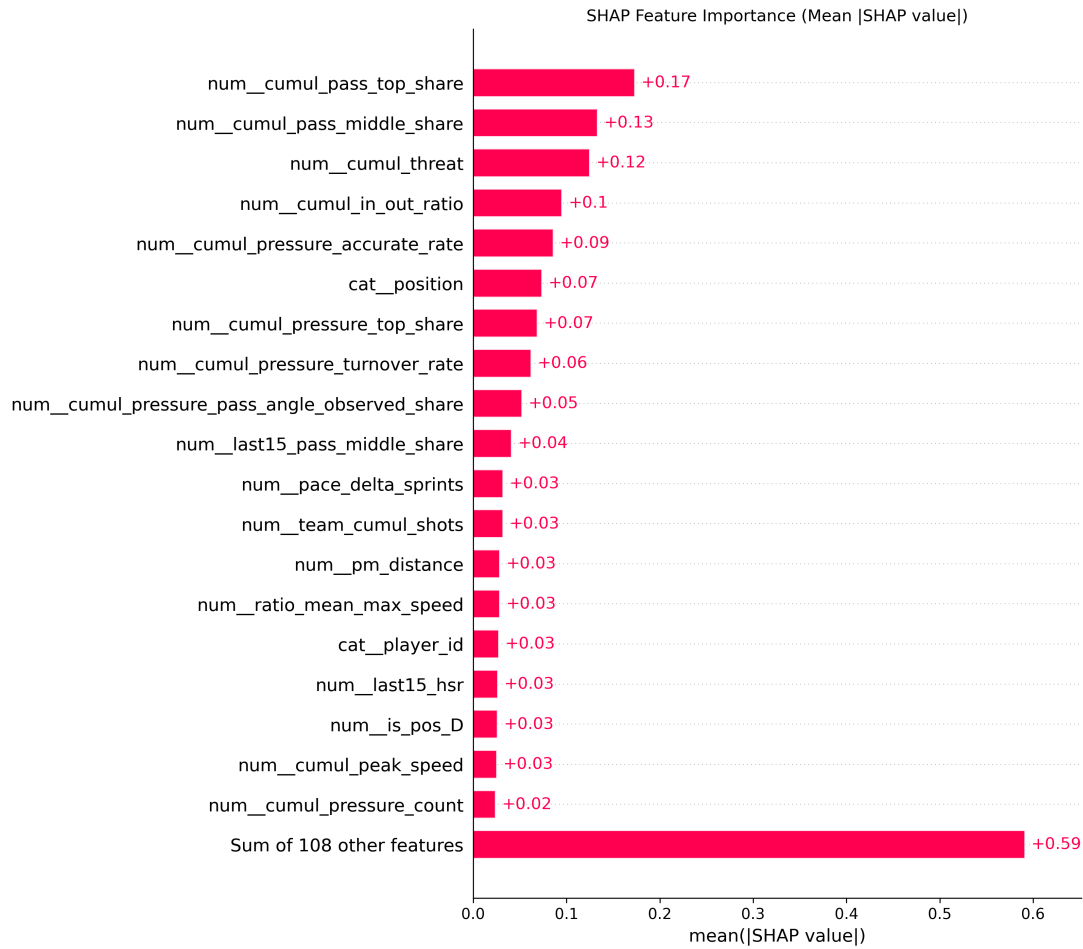


Figure 2. Top-10 SHAP feature importance for the final V6 XGBoost Poisson model.

7.2.5 Ablation Results

Full V6 Versus Reduced Subsets The ablation results provide some of the clearest empirical evidence in the entire project. The baseline full V6 model, using the complete 128-feature representation, achieved mean balanced accuracy 0.6719, mean PR AUC 0.1690, mean ROC AUC 0.7187, uncalibrated Brier score 0.0527, and calibrated Brier score 0.05254, with spline calibration selected as best in that main baseline comparison. Although this is already a strong result, it was systematically challenged by much smaller reduced models.

The Top 10 subset achieved balanced accuracy 0.6751, PR AUC 0.1933, ROC AUC 0.7343, uncalibrated Brier 0.0522, and calibrated Brier 0.05153 with isotonic calibration. This result is highly informative because it shows that a very small subset of the highest-ranked features already captures the sharpest ranking signal in the V6 space. The Top 20 subset improved balanced accuracy further to 0.6810, achieved PR AUC 0.1869, ROC AUC 0.7450, uncalibrated Brier 0.0526, and the best calibrated Brier score of 0.05129, again with isotonic calibration in the main comparison table. The Top 30 subset then slipped back to balanced accuracy 0.6739, PR AUC 0.1697, ROC AUC 0.7284, uncalibrated Brier 0.0537, and calibrated Brier 0.05262, with spline calibration selected as best in that comparison.

This pattern is difficult to interpret as anything other than strong evidence of overexpansion in the full V6 space. Once the model is compressed to a compact SHAP-guided subset, performance improves across nearly every core metric. Gains are strongest between the full baseline and the Top 10 or Top 20 subsets, and then flatten or partially regress once the subset expands toward Top 30. The information profile therefore looks like a clear plateau: the most valuable predictive content is concentrated very early in the feature ranking, and additional variables increasingly contribute variance rather than new signal.

Calibration Behavior The V6 line also produced a coherent calibration story. In the main comparison table, isotonic calibration emerged as the best calibration method for both the Top 10 and Top 20 subsets, whereas spline remained competitive and was best for the baseline full model and the Top 30 subset. For the finalized Top 20 run, the calibration script selected isotonic regression as the strongest method, reducing the Brier score from an uncalibrated 0.05254 to 0.05129. In the same comparison, spline delivered 0.05172, beta calibration 0.05203, and sigmoid calibration 0.05387.

This result matters because it suggests that the reduced feature spaces are not only more discriminative but also easier to calibrate reliably. Once the noise of the broader feature space is removed, the model appears to produce a smoother and more trustworthy mapping from intensity scores to event probabilities. For a paper that aims to connect prediction to interpretable match-risk assessment, this is a substantial practical advantage.

7.2.6 Interpretation of the V6 Findings

Dimensionality and Information Concentration The first major conclusion is that the V6 ablation study verifies a dimensionality problem in the practical modeling sense. The full 128-feature representation is not the strongest version of the model. Instead, reduced spaces built from the SHAP ranking perform better on balanced accuracy, ROC AUC, and calibrated Brier score, while the Top 10 subset also improves PR AUC dramatically. This indicates that the final stages of feature engineering had reached a point where adding more variables no longer improved prediction and instead introduced instability or noise.

Top 10 Efficiency Versus Top 20 Stability The second major conclusion is that the distinction between Top 10 and Top 20 is substantive rather than cosmetic. The Top 10 subset delivers the strongest PR AUC, which means that the most important rare-event ranking signal is captured very early in the feature ordering. However, the Top 20 subset provides the best balanced accuracy, the best ROC AUC, and the best calibrated Brier score. In other words, Top 10 is the most efficient ranking model, but Top 20 is the most balanced overall finalist. For a research context in which both classification stability and calibrated probabilities matter, the Top 20 subset is the more defensible final specification.

Implications for Football Interpretation The composition of the retained features also sharpens the football interpretation. The dominant variables are cumulative and structural rather than purely episodic. Passing shares across zones, accumulated threat, pressure-adjusted passing quality, passing centrality, and positional context suggest that later scoring probability emerges from persistent territorial and circulation patterns rather than only from isolated finishing events. This strengthens the paper’s broader claim that offensive output is embedded in how teams and players build, circulate, and sustain threatening states over time.

Poisson Objective as a Rare-Event Tool The final major conclusion concerns the modeling objective itself. The Poisson formulation remained effective throughout the V6 comparison, especially when combined with exposure normalization through the base margin. This means that the model did not merely treat scoring as a flat class label; it treated it as an event generated with varying intensity over varying opportunity lengths. That framing is statistically appropriate for football scoring and appears empirically robust in the V6 results.

7.2.7 Overall Takeaway

The V6 XGBoost Poisson phase offers one of the strongest and clearest messages in the paper. A Poisson-based XGBoost model with exposure normalization is a credible and effective way to treat later scoring as a rare-event intensity problem. However, the full engineered feature space is not the best final representation. The main predictive signal is concentrated in a compact SHAP-ranked subset. The Top 10 model captures the strongest ranking quality with PR AUC 0.1933, while the Top 20 model emerges as the most stable all-round finalist with balanced accuracy 0.6810, ROC AUC 0.7450, and the best calibrated Brier score of 0.05129. Overall, the V6 results show that disciplined feature compression, not maximal feature accumulation, is the key to extracting the strongest signal from the final engineered football representation.

7.3 K-Nearest Neighbors Modeling Results

The K-nearest neighbors line was developed as a deliberately simple and transparent benchmark built on the checkpoint-level dataset `data/knn_dataset.csv`. It was never expected to become the final contest-winning model. Instead, its role was methodological and scientific. First, it established a nonparametric baseline against which more structured learners could later be judged. Second, it tested whether local similarity in player-match states carries predictive information about later scoring. Third, it revealed how strongly the quality of a distance-based model depends on preprocessing choices such as feature scaling, removal of weak identifier-like variables, dimensionality reduction, and threshold selection. Fourth, it created a useful contrast with more expressive model classes, including tree ensembles and neural architectures, by showing what can be learned from local neighborhoods alone when the representation is limited to checkpoint-level summaries rather than richer sequential interaction

modeling.

7.3.1 Data Representation and Evaluation Protocol

The KNN experiments used `data/knn_dataset.csv`, a checkpoint-level dataset containing 3,486 rows and 31 columns. The target prevalence was approximately 5.82%, so the modeling task was a strongly imbalanced binary classification problem throughout. The feature space combined several types of information. Contextual variables included checkpoint timing, player position, formation, within-window minute, home status, and substitution status. Recent-performance summaries captured local behavior over the last fifteen minutes, including sprinting, high-speed running, shot activity, shots on target, shots under pressure, shots in the top third, and pass-volume and pass-accuracy aggregates. These variables were complemented by cumulative aggregates from the start of the match, such as cumulative sprints, cumulative high-speed running, cumulative shooting indicators, and cumulative passing summaries, together with an exposure variable measuring total minutes already played in the game.

The row identifier `scored_after_eval_key` was used to reconstruct a grouped evaluation structure resembling fixture-level separation, and model assessment relied on `StratifiedGroupKFold` with five folds. This was an appropriate evaluation design because it reduced the risk of leakage while preserving a reasonable degree of class-balance stability in a small rare-event setting. The primary comparison metric was precision-recall AUC, which is the most relevant ranking criterion for a low-prevalence target. Secondary metrics included balanced accuracy, ROC AUC, and Brier score, so the KNN line could be evaluated not only as a ranking model but also in terms of thresholded discrimination and probability quality.

7.3.2 Model Specification and Development Path

Baseline Specification The initial KNN pipeline used a leakage-safe preprocessing stack tailored to mixed tabular data. Numeric variables were median-imputed and standardized, categorical variables were imputed with their most frequent category and then one-hot encoded, and boolean variables were converted to integer form and imputed in the same most-frequent fashion. On top of this preprocessing layer, the classifier was a `KNeighborsClassifier` tuned over a compact but informative grid: `n_neighbors` in {5, 11, 21, 31}, `weights` in {uniform, distance}, and Minkowski distance with `p=1` for Manhattan distance or `p=2` for Euclidean distance. This design was intentionally modest. It was broad enough to test whether local geometry in the checkpoint representation had predictive value, yet compact enough to remain fully explainable in the paper as a clean benchmark rather than a heavily optimized search.

Original KNN on the Full Feature Space The first tuning stage kept the original feature set intact and asked the most basic question: can local similarity in the full checkpoint representation recover later-goal risk better than chance? The best tested specification at this stage used `n_neighbors=31`, `weights=distance`, and Manhattan distance, achieving mean PR AUC 0.0963 and mean ROC AUC 0.6152. This performance was modest, but it was still scientifically informative. It demonstrated that the checkpoint-level representation contained genuine local signal, because neighborhoods defined over recent performance, cumulative match behavior, and context were not entirely random with respect to later scoring. At the same time, the result was not strong enough to position KNN as a leading modeling family, which already suggested that the underlying football relationships were likely more interaction-heavy than what a simple distance vote could efficiently capture.

Removing ID-Like Features The next modeling step removed `player_id` and `jersey_number`. This was an important methodological correction rather than a cosmetic adjustment. In a distance-based model, arbitrary identifiers are particularly dangerous because they can distort the geometry of the feature space even when they do not represent meaningful football mechanisms. `player_id` is purely nominal, while `jersey_number` can function like a weak pseudo-identifier with little substantive relevance. After removing those variables and rerunning the compact search, the best reduced-feature configuration became `n_neighbors=31`, `weights=distance`, and Euclidean distance, with mean PR AUC 0.0967, mean ROC AUC 0.6251, mean balanced accuracy at threshold 0.5 equal to 0.5000, and mean Brier score 0.0555. The PR AUC improvement was small, but the ROC AUC gain was more noticeable, which indicates that the original neighborhood structure had indeed been partially polluted by ID-like coordinates. Removing those variables therefore improved methodological validity and made the local similarity space slightly more coherent.

Threshold Tuning on the Reduced-Feature KNN The reduced-feature KNN line then moved from pure ranking evaluation to threshold tuning on pooled out-of-fold probabilities. This was a crucial step in an imbalanced setting because a fixed threshold of 0.5 is typically too conservative when the positive class rate is below ten percent. KNN probabilities are also often only weakly calibrated in rare-event problems, so decision quality can depend heavily on the threshold even when the underlying ranking remains unchanged. The best reduced-feature threshold was 0.05. At that cutoff, the model achieved balanced accuracy 0.6098, recall 0.7143, precision 0.0820, and a predicted positive rate of approximately 0.50, with pooled out-of-fold PR AUC 0.0878 and pooled out-of-fold ROC AUC 0.5962. This result is important because it separates two notions of model usefulness. As a ranking model, the reduced-feature KNN remained only moderate. As a thresholded alerting model, however, it could be tuned into a high-recall system that captured many future scorers, albeit at the cost of low precision and a large number of false positives. In substantive terms, the KNN line showed that simple local

similarity can support recall-oriented screening, but not clean discrimination.

PCA Before KNN The final major extension inserted principal component analysis between preprocessing and distance calculation. This was designed to test whether latent compression could improve neighborhood behavior in a representation that mixed standardized numeric variables with one-hot encoded context and several correlated aggregates. KNN is known to struggle in sparse or high-dimensional spaces because distance becomes noisier and less behaviorally meaningful. The PCA search tested `n_components` in {8, 12, 16, 20, 24} together with the same compact KNN grid as before. The best PCA-based specification, ranked by mean PR AUC, used 12 components, `n_neighbors=31`, `weights=distance`, and Manhattan distance. It achieved mean PR AUC 0.1079, mean ROC AUC 0.6026, mean balanced accuracy at threshold 0.5 equal to 0.5000, and mean Brier score 0.0559.

This comparison is meaningful because the non-PCA reduced KNN achieved mean PR AUC 0.0967, whereas the PCA-enhanced version reached 0.1079. ROC AUC declined from 0.6251 to 0.6026, but in the present low-prevalence setting this trade-off is defensible. The PCA variant ranked positive cases earlier in the probability list, which is more relevant to the project's main predictive objective than preserving the exact ROC ordering. Threshold tuning for the PCA model selected a threshold of 0.08 and yielded balanced accuracy 0.5709, precision 0.0851, recall 0.4236, a predicted positive rate of 0.2900, pooled out-of-fold PR AUC 0.0878, and pooled out-of-fold ROC AUC 0.5962. Relative to the no-PCA threshold solution, the PCA version was more selective and less recall-heavy, illustrating once again how representation and operating threshold jointly determine the practical behavior of KNN.

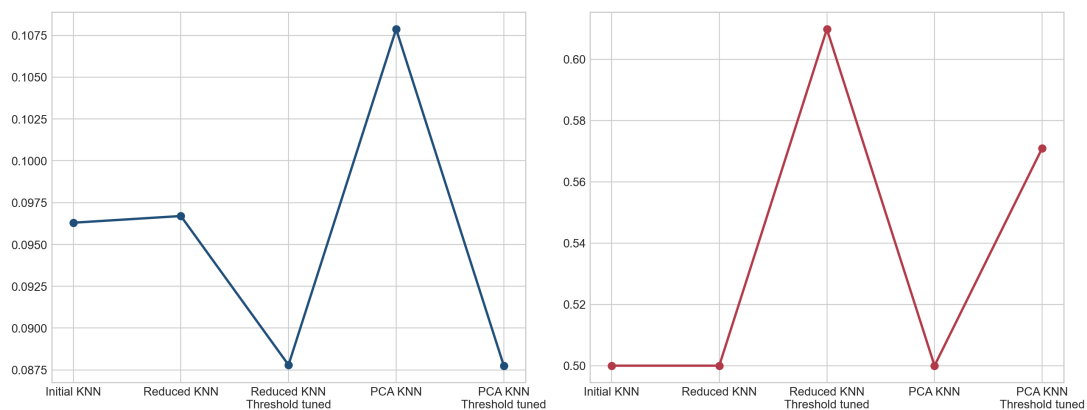


Figure 3. Evolution of KNN performance across successive modeling variants.

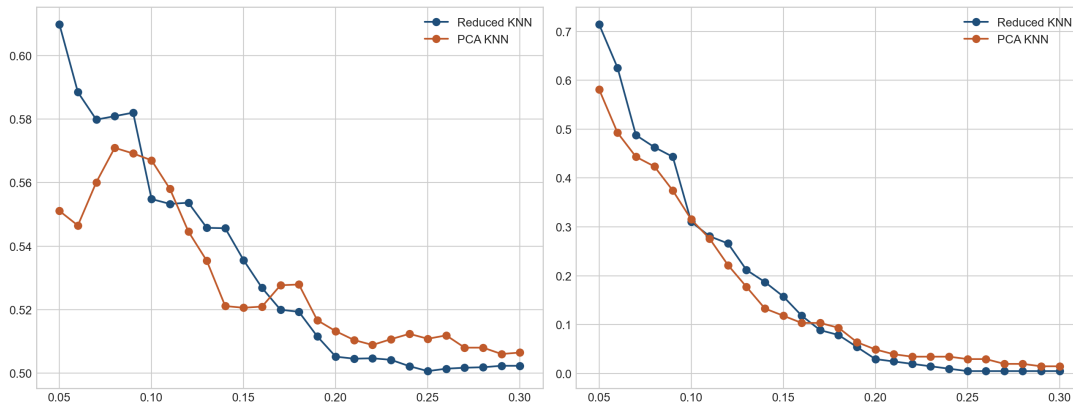


Figure 4. Threshold trade-offs for the reduced-feature and PCA-enhanced KNN variants.

7.3.3 Interpretation of the KNN Results

What KNN Shows About Predictive Signal The KNN family supports a restrained but useful conclusion about the project’s main predictive question. The strongest KNN ranking result was produced by the PCA-compressed model, which achieved mean PR AUC 0.1079. This is clearly above a trivial baseline and therefore supports the claim that checkpoint-level player behavior and match context contain real information about later scoring. However, the performance remains moderate, which indicates that local similarity alone is not the most efficient way to extract that information. In other words, football states that look alike in a wide tabular sense do share some later-goal risk, but the mapping from behavior to scoring is not simple enough to be fully captured by nearest-neighbor voting.

What KNN Suggests About Feature Structure KNN is not a strong tool for direct feature interpretation because it does not provide coefficients or stable native importance measures. Even so, its development path is informative. The improvement after removing `player_id` and `jersey_number` shows that meaningful geometry matters greatly in distance-based learning and that weak identifier-like variables can corrupt similarity more than they help. The PR AUC gain after PCA suggests that the useful football signal is distributed across correlated combinations of variables rather than concentrated in a few isolated axes. This is an indirect but important substantive insight: later scoring appears to depend on multivariate behavioral profiles, not merely on one or two simple raw indicators.

Implications for the Research Questions Relative to the broader research agenda, the KNN line contributes most strongly to the first question, namely whether player behavior can predict later scoring at all. The answer is yes, but only moderately well in a simple local-similarity framework. For the remaining questions, the KNN results are more suggestive than definitive. Because the dataset included recent and cumulative performance summaries, passing information, and contextual variables, the above-chance performance implies that all these sources of

information help define useful neighborhoods. However, KNN does not isolate their separate effects cleanly. It therefore cannot by itself identify whether passing is more important than shooting, whether short-term behavior dominates cumulative state, or which contextual factors matter most. Its value lies in showing that these elements collectively contain signal while also revealing that stronger nonlinear learners are needed for sharper substantive answers.

7.3.4 Discussion and Takeaway

The KNN family should therefore be interpreted as a methodological baseline rather than a final explanatory model. Its strengths are conceptual simplicity, fully nonparametric nonlinearity, and strong didactic value for demonstrating grouped cross-validation, threshold tuning, and the consequences of class imbalance. Its weaknesses are equally clear: limited interpretability at the feature level, high sensitivity to representation choices, vulnerability to sparse and high-dimensional spaces, and strong dependence on threshold selection when one wants usable classification behavior. The most defensible paper-level takeaway is that a simple distance-based learner can recover non-trivial signal from checkpoint-level football behavior, confirming that similar player-match states tend to share some later scoring risk. At the same time, its predictive quality remains modest, and its behavior changes materially when identifier-like variables are removed, when the feature space is compressed, and when the operating threshold is tuned. For that reason, KNN is best treated as a transparent baseline that justifies the move toward more expressive models rather than as a primary final candidate.

7.4 Sequence-Aware Neural Modeling Results

The sequence-aware modeling line was designed around the actual representation of the available data. Rather than starting from raw event-by-event tensors, the modeling pipeline used a checkpoint-level wide table in which sequential football behavior had already been summarized into structured families of features. This representation implied that the problem should be treated primarily as interaction-rich tabular learning with sequence-aware feature blocks, rather than as a classical recurrent-sequence problem. For that reason, the development path progressed in stages: first through tabular baselines, then through sequence-feature trimming, then through a custom dual-branch neural model, and finally through FT-transformer-style architectures with rare-event-oriented objectives.

7.4.1 Data Representation and Evaluation Protocol

The sequence-aware modeling line used a dedicated sequencing dataset consisting of 3,486 rows and 99 columns. The target prevalence remained approximately 5.82%, so the task continued to be a rare-event binary classification problem. After excluding identifiers and the target itself, the dataset contained 27 non-sequence context and aggregate columns,

24 `seq15m_*` features, 14 `seqpos_*` features, and 30 `seqextra_*` features. Conceptually, the `seq15m_*` family captured short-window timing, pressure, burstiness, and recency patterns; the `seqpos_*` family described possession linkage and escalation toward shots; and the `seqextra_*` family summarized additional composition variables such as play pattern, stage, body part, and technique structure.

The evaluation protocol remained fully aligned with the frozen grouped split logic used elsewhere in the project. Development folds and the holdout assignment were reconstructed from the existing evaluation metadata, so every experiment stayed consistent with the same fixture-level separation used in the rest of the modeling pipeline. Because the task is strongly imbalanced, the primary comparison metric was precision-recall AUC. Secondary metrics included balanced accuracy, ROC AUC, and Brier score. This evaluation setup is appropriate because the central contest question is not merely whether the model can classify outcomes, but whether it can rank rare scoring events effectively.

7.4.2 Model Development

Sequence-Aware Tabular Benchmark The first step asked whether richer sequence-aware features improved performance when the problem was kept in the wide checkpoint representation. The initial benchmark compared Logistic Regression, XGBoost, and a small multilayer perceptron over several feature sets, including `base_context`, `base_plus_sequence`, `full_engineered`, and trimmed subsets of `seq15m` and `seqpos`. This was a necessary first test because the input had already been aggregated, and the most practical question was whether more structured sequence summaries improved performance before increasing architectural complexity.

A key result of this stage was that naively adding the entire sequence block degraded performance. For XGBoost, the `base_context` feature set achieved a mean PR AUC of 0.1161, whereas both `base_plus_sequence` and `full_engineered` dropped to 0.0942. This result is substantively important because it shows that richer sequence-aware representation is not automatically useful: if the feature block is too broad, noisy, or redundant relative to the available sample size, the additional information can reduce predictive quality rather than improve it.

Trimming the Sequence Families Because the full sequence block proved too noisy, the next step used ablation-driven trimming. The retained `seq15m` features concentrated on pressure totals, turnover pressure, longest active streak, recency-weighted pressure, active-minute share, late bursts, and shot creation immediately after pressure. The retained `seqpos` features focused on strict link rate, shots per possession, shot-producing possessions, pressure-to-shot escalation, forward-pressure shot share, linked pressure volume, and run intensity per possession. This selection was guided by two principles: concentration of predictive signal in importance tables and clearer football interpretability than the noisier `seqextra_*` family.

The trimmed sequence results showed a more nuanced picture. Relative to the full unfiltered sequence block, trimming clearly improved the quality of the sequence-aware representation. However, even the best trimmed XGBoost variants still did not outperform `base_context` on the primary PR AUC metric. The combined trimmed specification reached mean PR AUC 0.1048, compared with 0.1161 for `base_context`, although it slightly improved balanced accuracy to 0.5595. This means that the sequence features were not useless; rather, they were directionally informative but not yet extracted efficiently by the baseline tabular architecture.

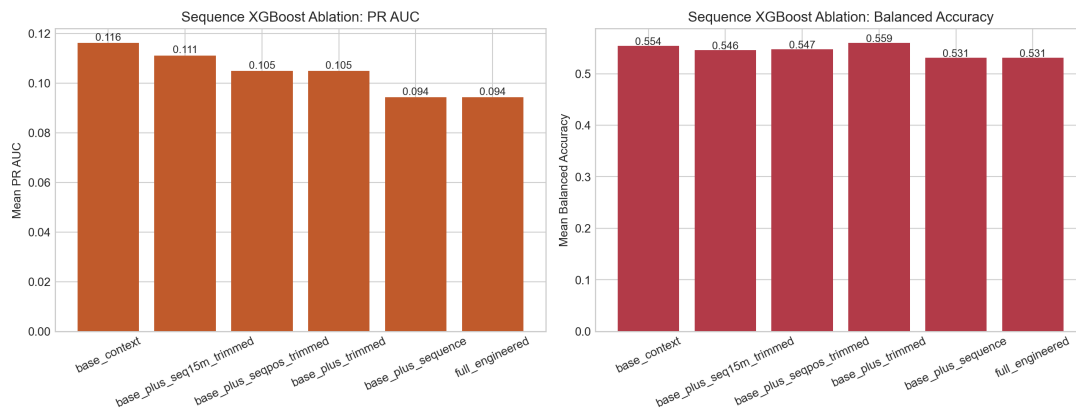


Figure 5. Sequence-aware XGBoost ablation results across feature-set variants.

Why Recurrent Models Were Rejected At this stage, a natural alternative would have been to move toward RNN, GRU, or LSTM architectures. That path was deliberately rejected. The reason was methodological: the available data no longer consisted of raw event-order tensors, but of checkpoint-level summaries in which much of the original temporal detail had already been collapsed into engineered features. A recurrent neural network would therefore inherit the cost and complexity of sequence modeling without receiving a genuinely sequential representation. This motivated a transition toward architectures that are natively stronger on mixed, interaction-rich tabular data.

Dual-Branch Neural Benchmark The next neural step was implemented through a dual-branch residual MLP. This design explicitly separated base context and trimmed sequence features into two streams: one branch processed core context plus categorical embeddings, the second processed the trimmed sequence block, and the two representations were fused only after separate transformation. The logic was straightforward: sequence information might still be useful, but only if it received its own representation path instead of being mixed naively with the full tabular set from the start.

This model did not surpass the strongest baselines on PR AUC, but it produced the best balanced-accuracy-oriented decision results after threshold tuning. In the neural comparison stage, the dual-branch residual MLP achieved mean PR AUC 0.1019, mean ROC AUC 0.6440,

and tuned out-of-fold balanced accuracy 0.6182. By contrast, the lighter FT-transformer-style model already reached a better mean PR AUC of 0.1210, outperforming both tree baselines and the custom MLP on the primary ranking objective. This was a crucial turning point in the modeling story: the dual-branch design improved class-balance behavior, but the transformer-style architecture improved the contest-relevant ranking metric.

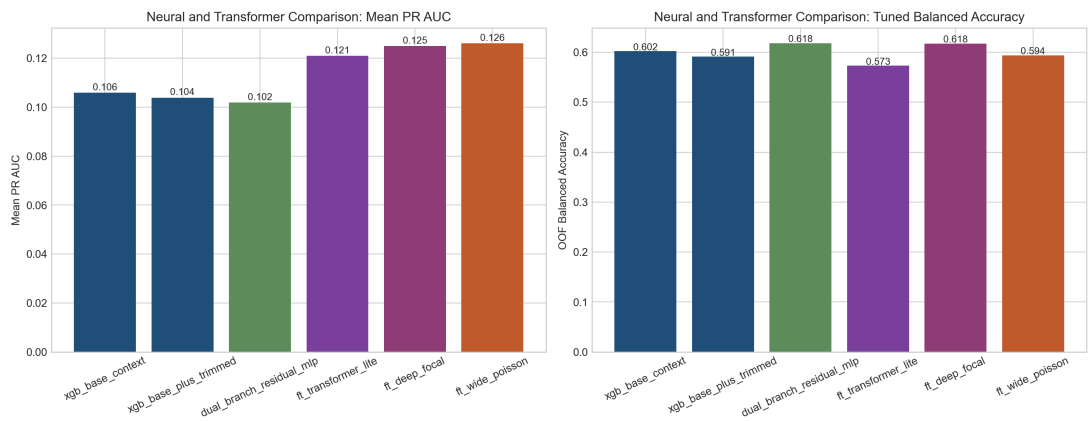


Figure 6. Comparison of benchmark neural architectures in the sequence-aware modeling line.

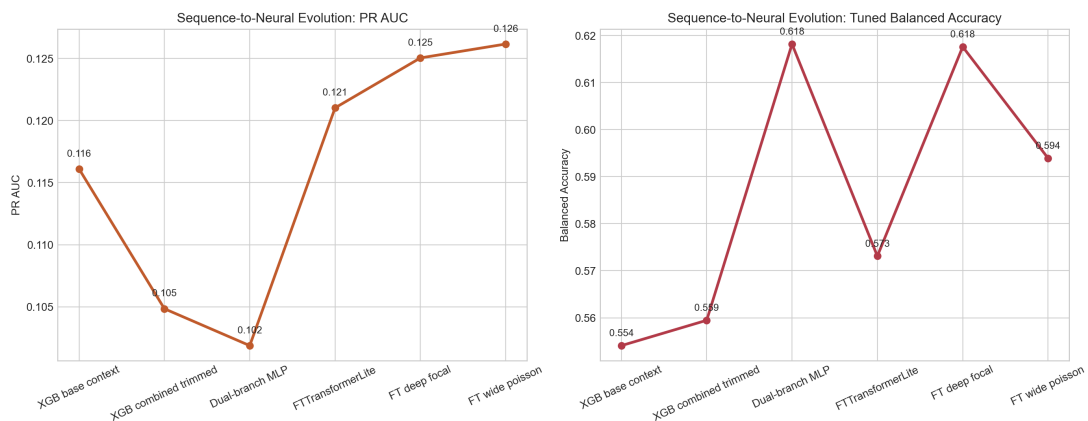


Figure 7. Performance evolution across the sequence-aware neural modeling stages.

Why the FT-Transformer Performed Better The FT-transformer family proved more suitable because it treats features as tokens rather than as one dense undifferentiated vector. This is particularly attractive in the present setting, where the signal likely depends on higher-order interactions between base context, recent pressure, possession linkage, and finishing context. In addition, the dataset contains a mixture of continuous aggregates, sparse contextual variables, and rare-event behavior; FT-style architectures are specifically designed for this kind of mixed tabular regime. Finally, because the target is highly imbalanced, the transformer line could be extended naturally with better rare-event objectives rather than relying only on a change in architecture.

FT-Transformer Search and Rare-Event Objectives The larger FT-transformer search systematically expanded token width, encoder depth, and attention heads, while also comparing multiple loss functions: binary cross-entropy, focal loss, and Poisson-style intensity loss. The Poisson framing was especially important conceptually. In this setup, the model predicts a latent scoring intensity parameter and the final reported goal probability is derived from that intensity. This preserves alignment with the contest question while allowing the internal model to learn later scoring as a rare-event process.

The completed search produced the strongest predictive result of the entire sequence-aware neural line. The best PR AUC was achieved by the `ft_wide_poisson` specification, which reached mean PR AUC 0.1262, mean ROC AUC 0.6359, and tuned out-of-fold balanced accuracy 0.5939. The best threshold-tuned balanced accuracy among the transformer-search models was delivered by `ft_deep_focal`, which reached 0.6176 while preserving substantially stronger ranking quality than the earlier dual-branch benchmark. These results support the broader modeling hypothesis: once the representation is compact and sequence-aware, the remaining gains come less from adding more engineered variables and more from learning richer feature interactions under rare-event-sensitive objectives.

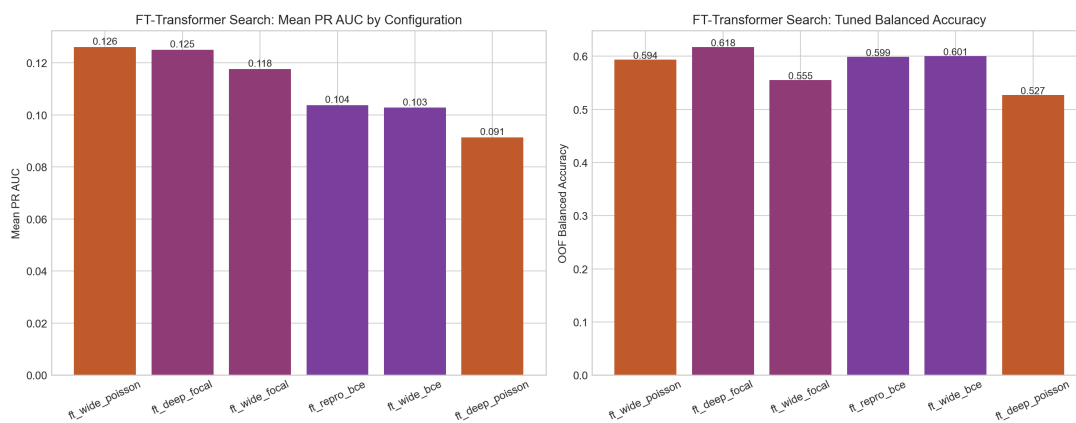


Figure 8. Comparison of FT-transformer search configurations and rare-event objectives.

7.4.3 Main Empirical Findings

Predictive Performance Taken together, the sequence-aware neural line gives the strongest answer in the project so far to the core predictive question. The best earlier sequence-stage XGBoost benchmark, using `base_context`, achieved mean PR AUC 0.1161. The first FT-transformer-style neural benchmark improved this to 0.1210. The best model in the completed transformer search, `ft_wide_poisson`, further improved the result to 0.1262. This progression is meaningful because it shows that player checkpoint behavior does contain predictive information about later scoring above trivial baselines, and that a tabular transformer with a rare-event-aware objective currently exploits that structure better than both the earlier tree benchmark and the lighter neural model.

Interpretation of Feature Groups The sequence-aware modeling line also strengthens the substantive interpretation of which mechanisms matter. The most credible retained features are not just raw counts of shots or sprints; instead, they describe pressure volume, pressure-induced turnovers, temporal clustering of activity, possession linkage, escalation toward shots, and forward-progressive attacking structure. This implies that the probability of later scoring depends not merely on isolated offensive actions, but on how recent pressure and possession structure interact to create shot-producing sequences. In that sense, the sequence-aware line contributes both predictive improvement and a richer football interpretation.

Short-Term Versus Cumulative Signals The retained feature families also suggest that recent local behavior matters strongly. Features such as recency-weighted pressure totals, active-minute share, late-burst counts, and longest active streak survive trimming more convincingly than many broader cumulative or composition-style features. This points toward an answer in favor of short-term local structure as an important determinant of later scoring probability. Cumulative state still matters, but it appears to be especially useful when interpreted alongside recent concentrated bursts of activity rather than as a substitute for them.

Role of Pressure, Passing, and Possession Context The results argue against a narrow interpretation in which sprinting and shooting variables alone are sufficient. Pressure-linked and possession-linked summaries repeatedly emerge as the most credible sequence-aware signals. However, the evidence also shows that such information is useful only when represented compactly and behaviorally, not when included as a large noisy block. Thus, passing context and pressure behavior do appear to matter, but only after careful feature selection or when modeled by an architecture capable of extracting interaction effects effectively.

External and Contextual Factors The neural modeling line also reinforces the role of contextual variables. Context was retained in every meaningful benchmark, and the strongest neural designs explicitly relied on interaction between context variables and recent-form summaries. This supports the conclusion that external factors such as match context, possession structure, and broader team-level circumstances continue to matter even when the model is enriched with sequence-aware summaries.

7.4.4 Discussion

The overall evidence supports a balanced conclusion. Sequence-aware features are behaviorally meaningful, but naive inclusion of the full engineered sequence block harms performance. Trimming improves their usefulness, but not enough on its own to beat the best context-only tree baseline on PR AUC. The first custom neural step, the dual-branch residual MLP, improved threshold-tuned classification balance but did not improve the main ranking objective. By contrast, the FT-transformer family consistently provided the strongest ranking

performance, and the Poisson-based variant currently gives the best executed PR AUC in the whole sequence-aware line. In other words, the main bottleneck is not whether sequence information matters, but how compactly it is represented and how effectively its interactions with context are modeled.

7.4.5 Limitations

Several limitations remain. First, the representation is still aggregated: even the neural models do not learn from raw event chains, but from checkpoint-level summaries. Second, the sample is still small for high-capacity neural modeling, with only 3,486 rows and a target prevalence of around 5.8%, so metric variance and instability remain relevant concerns. Third, although neural models improved the primary ranking metric, their probability calibration remains weaker than that of simpler baselines, as reflected in the Brier comparisons. This does not invalidate the models, but it means that any practical decision-support interpretation should treat their raw probability outputs with appropriate caution.

7.4.6 Overall Takeaway

The strongest concise interpretation of the results is that sequence-aware modeling improved the football realism of the representation, but only after strong compression of the feature families and the use of architectures designed for interaction-rich tabular data. The best executed model is currently the Poisson-based FT-transformer configuration `ft_wide_poisson`, which achieved mean PR AUC 0.1262. This result outperforms the earlier FT-transformer-lite benchmark and the earlier XGBoost sequence benchmark, while the focal-loss transformer variant delivered the strongest threshold-tuned balanced accuracy among the transformer-search models. Overall, the evidence indicates that compact pressure- and possession-oriented sequence summaries are useful, especially when modeled through architectures that can learn higher-order interactions and can treat later scoring as a rare-event intensity process rather than only as direct binary classification.

8 Hypothesis Verification

The empirical results confirm that players' in-game behavior contains meaningful predictive information about future goal-scoring probability. The baseline logistic regression already achieved a balanced accuracy of about 60%, indicating performance above random chance. More advanced models, particularly the XGBoost Poisson framework, further improved predictive power, supporting H1.

The analysis also demonstrates that different components of player behavior significantly influence scoring likelihood. Variables related to offensive actions (e.g., shots), passing structure, and positional context were statistically significant or highly ranked in feature importance

analyses, confirming H2 and H4. In contrast, models relying only on limited features such as sprinting or shooting underperformed compared to those using a broader feature set, supporting H3.

Evidence from both regression coefficients and SHAP-based feature importance indicates that contextual and cumulative variables, such as passing patterns, pressure-adjusted performance, and tactical role, play a crucial role, confirming H7. Additionally, short-term performance indicators (e.g., last 15 minutes activity) showed strong predictive relevance, providing partial support for H5 and H6. At the same time, their effect is best understood in combination with cumulative and contextual factors rather than as a uniformly stronger signal in isolation.

Overall, the hypotheses receive broad support: goal-scoring probability is not driven by a single metric but emerges from a combination of short-term intensity, accumulated actions, and tactical context.

9 Conclusions

This study demonstrates that predicting goal-scoring probability in football requires an integrated modeling approach combining physical, technical, and contextual data. Even relatively simple econometric models can capture non-random patterns, but machine learning approaches, especially XGBoost with a Poisson formulation, provide substantially better performance and interpretability.

A key finding is that predictive signal is highly concentrated in a limited number of well-designed features. Feature selection based on SHAP values showed that reduced models (Top 10–20 variables) outperform the full feature set, indicating that excessive feature expansion introduces noise rather than useful information.

From a football perspective, the results highlight that scoring probability is strongly linked to possession structure, passing dynamics, and accumulated attacking pressure, rather than isolated events like individual shots. This suggests that successful offensive play is a sustained process rather than a sequence of independent actions.

Finally, modeling goal-scoring as a rare-event intensity process (via Poisson-based methods) proved both theoretically appropriate and empirically effective. This approach not only improves prediction but also provides a more realistic interpretation of how scoring opportunities evolve over time.

In practical terms, the findings can support coaching decisions, substitution timing, and performance analysis by identifying when players or teams are most likely to create scoring opportunities.

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